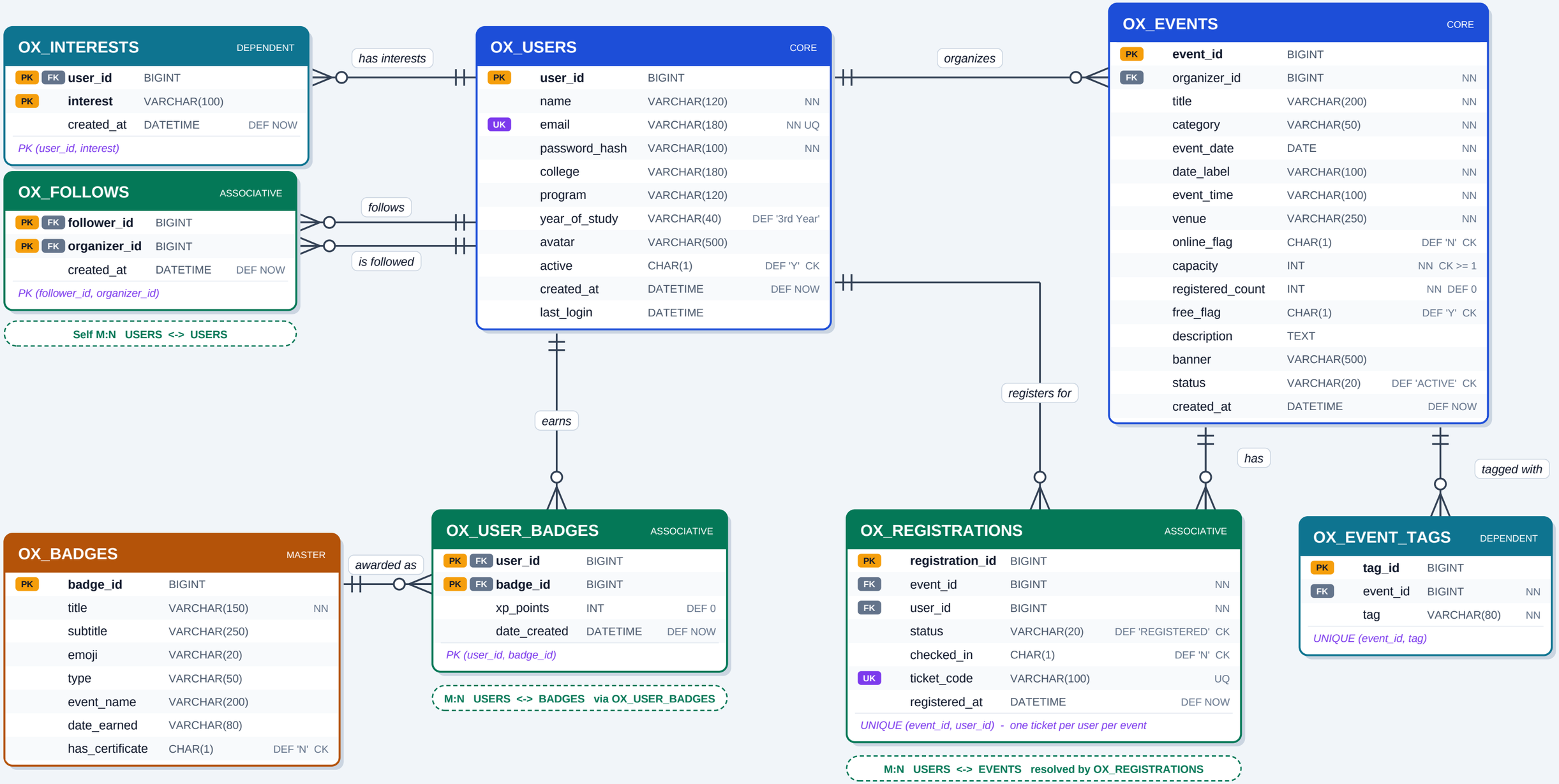
OpportuneX - Entity Relationship Diagram

Spring Boot + MySQL 8 + Stored Procedures | 8 tables, 9 relationships | generated from database/01_schema.sql



Legend

PK Primary key
FK Foreign key
UK Unique key

exactly one
zero or many

NN not null | UQ unique | CK check | DEF default

Core Bridge Dependent Master

Integrity rules (schema, triggers, procedures)

- One registration per user per event: UNIQUE (event_id, user_id); ticket_code is UNIQUE.
- Trigger ox_reg_ticket_trg fills ticket_code (OPX-<event>-<user>-<time>) if it is NULL.
- Trigger ox_events_capacity_trg blocks capacity below registered_count.
- ON DELETE CASCADE: tags, registrations, follows, interests, user_badges (user side).
- No cascade (restrict): ox_events.organizer_id and ox_user_badges.badge_id.
- Events are soft-cancelled: status = 'CANCELLED' (ox_cancel_event).
- registered_count is a denormalised seat counter kept in sync by ox_toggle_registration.
- All Y/N flags and status columns carry CHECK constraints; capacity >= 1.

Auto-increment, indexes and procedures

AUTO_INC	users from 100, events from 2000; tags, registrations, badges from 1
Indexes	ix_ox_events_date, ix_ox_events_org, ix_ox_reg_user, ix_ox_follows_org, ix_ox_user_badges_badge + unique keys on email, ticket_code, (event, user)
Procedures	ox_create_user, ox_update_profile, ox_add/remove_interest, ox_toggle_follow, ox_create/update/cancel_event, ox_save_tags, ox_toggle_registration, ox_checkin_ticket, ox_toggle_attendee
Errors	SIGNAL SQLSTATE 45001..45005 -> HTTP 404, 409, 400, 403, 409

Relationships and Data Dictionary

Every foreign key is NOT NULL, so each child row always has a parent; a parent may have zero or many children.

Relationships

#	Relationship	Parent (1) -> Child (N)	Foreign key	On delete	Business meaning
1	**organizes**	OX_USERS -> OX_EVENTS	ox_events.organizer_id	NO ACTION	A user hosts many events; every event has exactly one organizer.
2	**tagged with**	OX_EVENTS -> OX_EVENT_TAGS	ox_event_tags.event_id	CASCADE	An event has many tags; tags disappear with the event.
3	**registers for**	OX_USERS -> OX_REGISTRATIONS	ox_registrations.user_id	CASCADE	A user can register for many events (ticket per registration).
4	**has registrations**	OX_EVENTS -> OX_REGISTRATIONS	ox_registrations.event_id	CASCADE	An event collects many registrations, capped by capacity.
5	**has interests**	OX_USERS -> OX_INTERESTS	ox_interests.user_id	CASCADE	A user keeps many interests; PK (user_id, interest) blocks duplicates.
6	**follows**	OX_USERS -> OX_FOLLOWS	ox_follows.follower_id	CASCADE	The follower side of a follow; user can follow many organizers.
7	**is followed**	OX_USERS -> OX_FOLLOWS	ox_follows.organizer_id	CASCADE	The organizer side of a follow; an organizer has many followers.
8	**earns**	OX_USERS -> OX_USER_BADGES	ox_user_badges.user_id	CASCADE	A user can earn many badges, each with its own XP.
9	**awarded as**	OX_BADGES -> OX_USER_BADGES	ox_user_badges.badge_id	NO ACTION	One badge definition can be awarded to many users.

Entity summary

Entity	Type	Primary key	Foreign keys	Unique / check	Purpose
OX_USERS	Core	user_id	-	email UNIQUE; active IN (Y,N)	Accounts for students and organizers (BCrypt password hash, last login).
OX_EVENTS	Core	event_id	organizer_id	capacity >= 1; status IN (ACTIVE, CANCELLED); flags Y/N	Hackathons, workshops, seminars, cultural fests and competitions.
OX_EVENT_TAGS	Dependent	tag_id	event_id	-	Searchable tags per event.
OX_REGISTRATIONS	Bridge	registration_id	event_id, user_id	UNIQUE (event_id, user_id); ticket_code UNIQUE; status IN (REGISTERED, CANCELLED); checked_in Y/N	Resolves M:N Users and Events; holds ticket code and check-in state.
OX_FOLLOWS	Bridge	(follower_id, organizer_id)	follower_id, organizer_id	composite PK	Self-referencing M:N: users following organizer accounts.
OX_INTERESTS	Dependent	(user_id, interest)	user_id	composite PK	Interest keywords used for personalised discovery.
OX_BADGES	Master	badge_id	-	has_certificate IN (Y,N)	Badge / certificate definitions (title, emoji, event name, type).
OX_USER_BADGES	Bridge	(user_id, badge_id)	user_id, badge_id	composite PK	Resolves M:N Users and Badges; stores XP earned per badge.